Total No. of Questions : 8]

SEAT No. :

PC2484

[Total No. of Pages : 4

[6354]-613

B.E. (Mehanical Engineering)

DYNAMICS OF MACHINERY

(2019 Pattern) (Semester-VII) (402042)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4,Q.5 or Q.6 Q.7, or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Use of logarithmic tables, slide rule, and electronic pocket calculator is allowed.*
4) *Figures to the right indicate full marks.*
5) *Assume suitable data, if necessary.*

Q1) a) Find the natural frequency of the system as shown in figure.1 using Energy Method or Newton's Approach. **[8]**

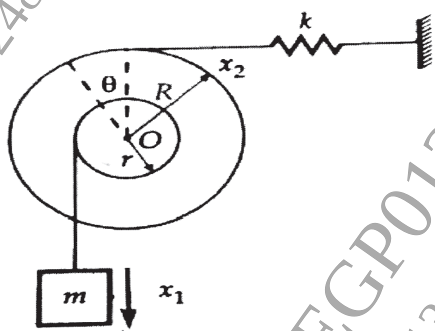

Figure. 1

b) Explain with a neat diagram mathematical model of a Bicycle. **[5]**

c) Explain the following terms used in vibration. **[4]**

i) Free Vibrations

ii) Forced Vibrations

OR

Q2) a) A body of 5kg is supported on a spring of stiffness 1960 N/m & has a dashpot connected to it, which produces a resistance of 1. 96 N at a velocity of 1 m/sec. In what ratio will be amplitude of vibration reduced after 5 cycles? **[8]**

b) Discuss over-damped, under-damped and critically damped system with the help of amplitude versus time plot. **[5]**

c) Define the following terms **[4]**

i) Viscous damping

ii) Coulomb damping

P.T.O.

Q3) a) A system having rotating unbalance has a total mass of 25 kg. The unbalanced mass of 1 kg rotates with a radius of 0.04m It has been observed that at a speed of 1000 rpm, the system & eccentric mass has a phase difference of 90 degrees and corresponding amplitude is 0.015 m. Find out **[8]**

i) natural frequency of the system

ii) damping factor

iii) amplitude at 1500 rpm

iv) Phase angle at 1500 rpm.

b) Explain frequency response curves (Magnification factor Vs. Frequency ratio) forced vibration due to external harmonic excitation. **[5]**

c) Define Quality factor and state its significance in frequency response curves. **[5]**

OR

Q4) a) The springs of an automobile trailer are compressed 0.1m under its own weight. Find the critical speed when the trailer is passing over a road with a profile of sine-wave whose amplitude is 80 mm and the wavelength is 14 m. Find the amplitude of vibration at a speed of 60 km/hr. **[8]**

b) What is critical speed of rotor without damping? Explain with the neat sketch. **[5]**

c) Explain Force transmissibility on the basis of **[5]**

i) Definition

ii) Equation

iii) Terminology used in equation

Q5) a) Figure 2, shows a vibrating system having two degrees of freedom. Determine the two natural frequencies of vibrations of motion of m_1 and m_2 for the two modes of vibration. Take m_1 = 20 kg, m_2 = 35 kg, K= 3000 N/m. **[10]**

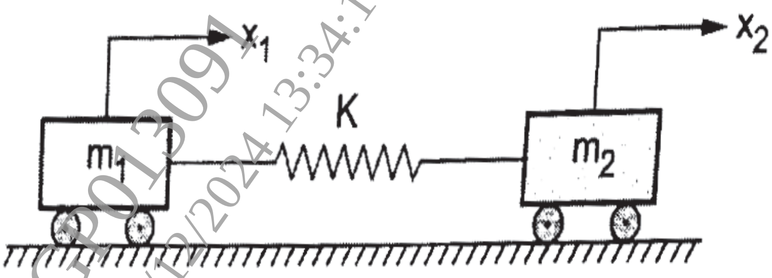

Figure. 2

b) Derive the equation for the length of Torsionally Equivalent Shaft. **[8]**

OR

Q6) a) Determine the natural frequencies and position of node of torsional vibration system having two rotors A and B attached to the ends of a shaft 1500 mm long. The MOI of rotor A is 650 kgm^2 and that of rotor B is 215 kgm^2. The shaft is 95 mm diameter for the first 600 mm, 60 mm diameter for the next 500 mm lenth and 50 mm diameter for the remaining length. Modulus of rigidity of shaft is 0.8 x 10^5 Mpa. (Assume value of equivalent diameter as 0.1 m). **[10]**

b) Explain free vibrations of a two rotor system using following parameters. **[8]**

i) Zero frequency

ii) Node point

iii) Position of node

iv) Amplitude ratios of two rotors

Q7) a) Differentiate Time domain and frequency domain Analysis. Explain how frequency spectrum can be used to detect vibration related faults in a system. **[8]**

b) Explain any one frequency measuring instrument with a neat diagram. **[5]**

c) Classify vibration measuring instruments. **[4]**

OR

Q8) a) Show that as a distance from a point source doubles, the Sound Intensity Level decreases by 6 dB. Assume that sound propagates in the form of spherical waves. **[8]**

b) Write a short note on Sound fields. **[5]**

c) Define the following term. **[4]**

i) Sound pressure level

ii) Sound Power level

iii) Sound intensity level

iv) Sound absorption coefficent

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